

MUHAMMAD FAIZAN AHMED

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Summary

MSc Robotics & Automation student at Manchester Metropolitan University with hands-on experience in PLC programming, SCADA development, and industrial automation system integration. Experienced in designing and implementing control and monitoring solutions for production environments. Strong Interest in robotics and automation engineering roles involving system design, integration, testing, and industrial deployment.

Skills

- **Industrial Automation & Control:** PLC Programming (Siemens, Delta), SCADA (WinCC Unified), HMI Development, Virtual Commissioning
- **Robotics, Vision & Smart Manufacturing:** Industrial Robotics, Cobot Coordination, Robot Simulation (RoboDK, CoppeliaSim), Machine Vision (OpenCV, YOLO), Industrial Inspection, IIoT Monitoring
- **Programming & Engineering Development:** Python, C++, MATLAB, Embedded Systems, PCB Design, 3D Modelling (SolidWorks, Fusion 360), Additive Manufacturing
- **Professional:** Technical Documentation, Problem Solving, Team Collaboration, Project Ownership

Professional Experience

Automation Engineer

Industronics Engineering | Mar 2025 – Aug 2025

- Designed and implemented a full WinCC Unified SCADA system for a fully automated warehouse for a UK-based client.
- Designed and deployed a vision-based cloth width measurement system using computer vision for real-time industrial textile inspection.
- Applied industrial control, machine vision, and system integration skills in production-focused automation projects.

Hardware Design Engineer

INTEL Lab, PNEC NUST | Mar 2024 – Mar 2025

- Developed GUIs using Python and worked with Linux systems for software integration.
- Designed PCBs from schematic to 3D models using Altium and EasyEDA, with 3D modelling and custom enclosure design using SolidWorks.

IoT Engineer

Indus Technetronic | Jul 2023 – Dec 2023

- Developed a home automation smart switching IoT device ("Beetle") for a startup, supporting hardware design and system integration.
- Worked on a residential load consumption monitoring system, enabling real-time energy usage analysis.

Education

MSc Robotics & Automation

Manchester Metropolitan University | Sep 2025 – Sep 2026 (Expected)

- **MSc Project – Cobot Synchronisation:** Coordinated control of collaborative robots for industrial sewing task optimisation.

Bachelor of Engineering in Mechatronics Engineering

Mehran University of Engineering & Technology | Oct 2019 – Dec 2023